GOWDUL AALAM M

Agentic AI Engineer | LangGraph, RAG & LLM Orchestration | Remote

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PROFESSIONAL SUMMARY

AI engineer with **5 years** designing production-grade agentic multi-agent systems (**LangGraph, LangChain, CrewAI**) and RAG pipelines for Fortune 500 clients (Verizon, Wells Fargo) processing **10M+ events/day** at **<200ms P95 latency**. Specialize in tool-calling orchestration, LLM evaluation & observability (**LangSmith, RAGAS**), and token-cost optimization — shipping systems on AWS, Azure, and GCP that cut manual effort, inference spend, and time-to-resolution at enterprise scale.

FEATURED PROJECTS

Agentic RAG Platform with Multi-Agent Evaluation & Observability

Stack: LangGraph · LangChain · FastAPI · FAISS/Pinecone · LangSmith · RAGAS · AWS Bedrock · Azure OpenAI · GCP Vertex AI

- Self-correcting retrieval: built a corrective-RAG loop in **LangGraph** where a critic agent scores retrieval confidence and triggers re-query/re-rank on low grounding, lifting answer faithfulness from **78% to 94%** on a 500-query eval set.
- Automated evaluation harness: integrated **RAGAS** to continuously score faithfulness, context precision, and answer relevancy in CI, gating deployments below a **90% faithfulness threshold** and catching regressions pre-release.
- Observability & cost control: instrumented end-to-end **LangSmith** tracing with custom token-cost dashboards; cut per-query token consumption **~35%** via prompt compression, **semantic caching**, and dynamic context trimming.
- Multi-cloud deployment: containerized on **AWS (ECS/Lambda)** with parity testing on **Azure OpenAI** and **GCP Vertex AI** endpoints for client-side portability requirements.

Multi-Agent Tool-Calling Orchestrator with Cost-Aware Model Routing

Stack: LangGraph · CrewAI · Model Context Protocol (MCP) · Pydantic · Redis · LangSmith

- Planner→Executor→Verifier graph: dynamic tool selection across **15+ MCP-registered tools** with schema-validated **Pydantic** outputs, reducing tool-call errors **~90%** vs. unconstrained function calling.
- Cost-aware model routing: built a router that switches between frontier and smaller models by subtask complexity, cutting inference cost **~40%** with no measurable drop in task success rate.
- HITL & audit trail: approval gates on high-stakes tool calls (writes, transactions) with **100% action traceability**, mirroring production compliance requirements.
- Scale-tested: **50+ concurrent agent sessions** at **sub-2s P95 planning latency** under load.

PROFESSIONAL EXPERIENCE

Senior Analyst — Generative AI & Automation | Accenture *Dec 2024 – Present*

Coimbatore, India

- Designed an **NLP-powered Call Centre Analytics** system ingesting JSON feeds from Verizon and Wells Fargo — performing root-cause detection, pattern mining, escalation identification, and callback intent extraction using LLM-based pipelines.
- Built interactive analytics dashboards surfacing recurring trends; improved issue resolution speed by **40%** and cut root-cause discovery time by **40%**.

- Benchmarked a PDF summarization tool across multiple **LLMs** (evaluating accuracy, coherence, latency, and cost), selecting the optimal model and achieving a **35% reduction** in processing time with improved output quality.
- Developed automated API testing POCs using **Karate Labs** with multi-step API chaining, reducing manual testing effort by **~40%** and improving end-to-end coverage.

SDE II — Backend & Integration Engineer | Mindgate Solutions *Apr 2024 – Nov 2024*
Chennai, India

- Integrated Middle Eastern banking bill payment providers (**Efawateer, Sadad**) to deliver secure payment solutions for corporate clients via robust API integrations.
- Collaborated with core banking teams to resolve transaction failures, reducing processing errors and improving throughput by **~20%**.

Software Developer — AI & Backend Engineering | Senseforth.ai Solutions *Jun 2021 – Jan 2024*
Bangalore, India

- Architected an **AWS Kinesis Firehose** streaming pipeline to S3, with automated ingestion via **ELK Stack (Logstash)**, eliminating manual uploads and improving data ingestion latency by **~50%** — successfully promoted from POC to production.
- Implemented LLM-powered **Text-to-Speech (TTS)** capabilities using Python and Java integrations, increasing accessibility and boosting user engagement by **30%** in a chatbot application.
- Built end-to-end **Salesforce–WhatsApp integration** using **Twilio APIs** (POC to production), automating messaging workflows, lead notifications, and two-way customer interactions via webhook handling.
- Enforced a consumption rate limit of **100 requests/day** on API services, reducing server load by **40%** — demonstrating LLM API cost optimisation and observability.

SKILLS

Agentic AI & Orchestration: LangGraph (StateGraph, checkpointing, conditional edges), LangChain, CrewAI, ReAct, Model Context Protocol (MCP), tool calling, multi-agent handoffs, Pydantic structured outputs

RAG & Retrieval: FAISS, Pinecone, ChromaDB, hybrid/semantic retrieval, HyDE, corrective retrieval, chunking strategies, semantic caching, sentence-transformer embeddings

Evaluation & Observability: LangSmith, RAGAS, prompt-version management, trace/latency profiling, eval-set gating, token-cost tracking & optimization

Cloud (as required): AWS (Bedrock, Kinesis Firehose, S3, EC2, Lambda), Azure (Azure OpenAI, Azure Functions), GCP (Vertex AI)

Core Languages & Frameworks: Python, Java, SQL, FastAPI, Spring Boot, React, PostgreSQL

DevOps / Tools: Docker, Git, CI/CD, ELK Stack, Kafka, Redis

EDUCATION

B.E. — Computer Science & Engineering | Karpagam College of Engineering | 2018 – 2022